

SEATTLE PAINE, WA, USA

WMO: 727937

Lat: 47.908N

Lon: 122.280W

Elev: 606

StdP: 14.38

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 24222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.4	29.0	11.8	10.3	30.3	17.5	13.6	32.5	28.8	49.0	26.0	47.7	4.9	120	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.3	80.9	63.8	77.2	62.3	73.8	61.1	65.2	77.9	63.6	75.0	62.2	72.3	6.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	81.1	68.4	58.8	75.7	67.3	57.3	71.8	65.9	30.3	77.7	29.1	75.1	28.1	72.1	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.4	20.1	17.4	21.0	87.5	4.8	4.1	17.5	90.4	14.7	92.9	12.0	95.2	8.5	98.2		
			19.0	68.0	4.6	1.7	15.6	69.2	12.9	70.2	10.4	71.1	7.0	72.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	41.3	42.0	45.2	49.0	54.5	58.9	63.5	64.3	59.8	51.9	45.0	40.2
	DBStd	9.80	5.81	5.33	4.95	4.83	5.11	5.02	4.57	3.99	4.45	4.46	5.78	5.68
	HDD50	1254	273	227	165	73	14	1	0	0	0	30	167	304
	HDD65	5088	735	645	615	481	328	197	83	61	168	406	601	768
	CDD50	1745	4	2	15	43	153	267	418	443	293	89	16	2
	CDD65	101	0	0	0	0	3	13	36	39	10	0	0	0
	CDH74	708	0	0	1	4	23	92	278	250	60	0	0	0
	CDH80	132	0	0	0	0	3	13	66	44	6	0	0	0
Wind	WSAvg	7.1	8.2	7.7	8.7	7.6	6.5	6.1	5.9	5.6	5.6	6.7	8.0	8.0
Precipitation	PrecAvg	35.2	4.5	3.0	3.9	2.4	2.2	1.7	0.6	0.9	1.7	3.5	5.4	5.3
	PrecMax	43.7	8.9	8.4	6.9	4.6	3.5	3.5	1.4	2.8	3.9	7.2	10.6	9.3
	PrecMin	26.3	2.2	1.3	2.1	0.5	0.5	0.4	0.0	0.0	0.4	0.7	2.4	1.8
	PrecStd	5.4	1.7	1.6	1.4	1.0	0.9	0.9	0.4	0.8	0.9	1.7	2.3	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	57.4	64.7	72.3	77.0	81.8	86.4	85.0	79.9	68.4	62.0	55.2
		MCWB	51.7	46.3	51.7	56.8	60.0	64.2	67.1	65.0	62.2	57.3	54.4	51.0
	2%	DB	53.8	54.0	58.9	64.3	71.0	76.6	80.9	80.2	75.0	63.8	57.2	52.0
		MCWB	49.8	46.5	49.0	52.8	58.4	62.1	64.4	64.0	60.8	55.4	52.7	48.3
	5%	DB	51.3	51.5	54.9	60.9	66.7	72.0	76.5	76.4	71.2	61.2	54.7	49.5
		MCWB	47.1	45.9	47.6	50.9	56.5	59.7	62.5	62.6	59.5	54.5	51.0	45.8
	10%	DB	48.9	49.1	52.4	56.7	63.2	68.0	72.8	73.0	67.8	58.5	52.6	47.6
		MCWB	45.2	44.7	46.7	48.8	54.7	57.9	61.1	61.3	58.4	53.2	49.2	44.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	50.8	53.6	57.3	61.8	66.0	68.4	66.5	64.7	60.3	56.6	53.2
		MCDB	55.4	53.3	60.2	69.3	73.4	79.7	83.8	80.3	75.8	65.2	59.4	54.8
	2%	WB	50.7	48.8	50.9	54.1	59.2	62.6	65.4	65.0	62.4	57.5	53.8	49.1
		MCDB	52.8	51.9	56.4	63.2	69.4	74.4	78.3	77.3	71.5	61.1	56.5	50.8
	5%	WB	47.9	46.7	49.2	51.8	57.3	60.4	63.4	63.6	60.7	55.7	51.8	46.5
		MCDB	49.9	49.6	53.9	58.3	65.3	70.1	75.3	74.6	68.6	59.3	54.4	48.1
	10%	WB	46.0	45.5	47.3	50.0	55.2	58.8	61.8	62.2	59.3	54.0	49.7	45.1
		MCDB	48.0	48.2	51.7	55.6	62.3	66.9	72.2	71.8	66.0	57.7	52.3	46.9
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.4	11.0	12.4	13.5	14.0	16.6	16.3	13.6	10.4	8.6	7.6
		MCDBR	10.1	12.7	15.0	18.3	19.6	20.6	22.4	21.6	19.5	14.1	10.4	9.3
		MCWBR	7.2	7.9	8.4	9.7	10.3	10.2	9.8	9.3	8.6	7.8	7.1	7.0
	5% WB	MCDBR	8.8	9.7	13.0	16.2	17.5	18.5	20.9	19.4	16.5	11.4	9.2	8.0
		MCWBR	7.3	7.1	8.1	9.3	9.8	9.9	9.8	9.0	8.0	7.7	7.3	7.1
Clear-Sky Solar Irradiance	taub	0.313	0.305	0.316	0.343	0.361	0.349	0.340	0.345	0.333	0.320	0.309	0.314	
	taud	2.527	2.565	2.558	2.461	2.428	2.481	2.530	2.523	2.548	2.584	2.565	2.461	
	Ebn at Noon	245	276	290	289	286	289	291	285	279	266	246	226	
	Edh at Noon	20	23	27	33	35	33	31	30	27	22	19	18	
All-Sky Solar Radiation	RadAvg	314	593	902	1341	1693	1859	2071	1759	1267	712	368	253	
	RadStd	44	77	104	125	136	179	151	116	109	80	51	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.86	-1.98	-4.00	+2.66	N/A	N/A	N/A	-286	+283	+59
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page